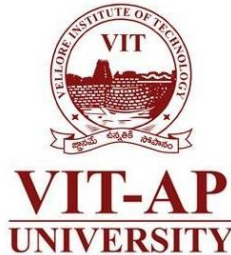


Differentiation of Integrals Containing Parameter: Leibniz's Rule



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In applications, we sometimes encounter functions like

$$f(x) = \int_{\sin x}^{x^2} (1 + t) dt \quad \text{and} \quad g(x) = \int_{\sqrt{x}}^{2\sqrt{x}} \sin t^2 dt,$$

defined by integrals that have variable upper limits of integration and variable lower limits of integration at the same time.

Leibniz's Rule

If f is continuous on $[a, b]$ and if $u(x)$ and $v(x)$ are differentiable functions of x whose values lie in $[a, b]$, then

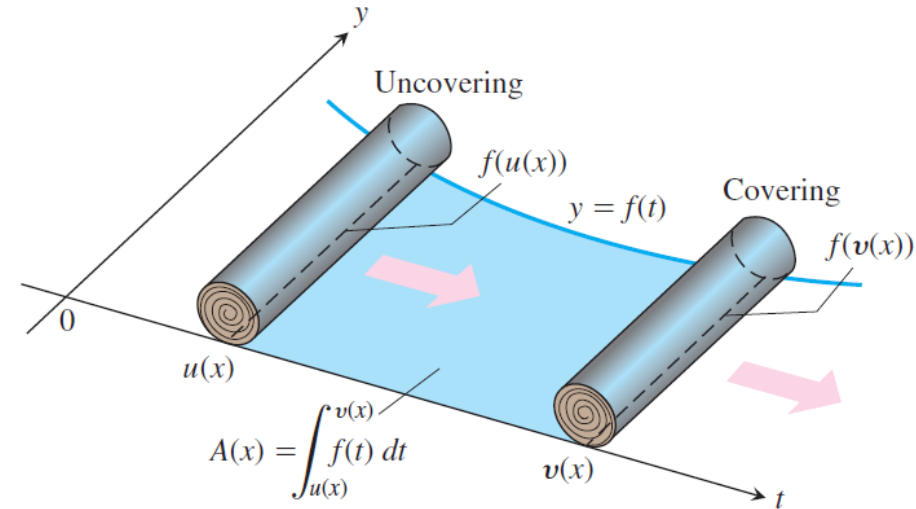
$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x)) \frac{dv}{dx} - f(u(x)) \frac{du}{dx}.$$

Leibniz's Rule - Interpretation

Figure gives a geometric interpretation of Leibniz's Rule. It shows a carpet of variable width $f(t)$ that is being rolled up at the left at the same time x as it is being unrolled at the right. (In this interpretation, time is x , not t .) At time x , the floor is covered from $u(x)$ to $v(x)$. The rate du/dx at which the carpet is being rolled up need not be the same as the rate dv/dx at which the carpet is being laid down. At any given time x , the area covered by carpet is

$$A(x) = \int_{u(x)}^{v(x)} f(t) dt.$$

$$\frac{dA}{dx} = f(v(x)) \frac{dv}{dx} - f(u(x)) \frac{du}{dx}.$$



Example

Problem-1

A carpet of variable width $\sin t^2$ that is being rolled up at the left at the same time x as it is being (time variable x not t). At time x , the floor is covered from $u(x) = \sqrt{x}$ to $v(x) = 2\sqrt{x}$. Assuming the rate $\frac{du}{dx}$ at which the carpet is rolled up is not the same rate $\frac{dv}{dx}$ at which the carpet is unrolled. Find the net rate change of the area covered by the carpet.

Hint: $A(x) = \int_{\sqrt{x}}^{2\sqrt{x}} \sin t^2 dt.$

$$\implies \frac{dA}{dx} = (\sin(\sqrt{x})^2) \frac{d\sqrt{x}}{dx} - (\sin(2\sqrt{x})^2) \frac{d(2\sqrt{x})}{dx}$$

$$\implies A'(x) = \frac{\sin 4x}{\sqrt{x}} - \frac{\sin x}{2\sqrt{x}}$$

Problem-2

A carpet of variable width $t(5 - t)$ that is being rolled up at the left at the same time x as it is being (time variable x not t). At time x , the floor is covered from $u(x) = x$ to $v(x) = x + 3$. Assuming the rate $\frac{du}{dx}$ at which the carpet is rolled up is not the same rate $\frac{dv}{dx}$ at which the carpet is unrolled. What is the maximum area covered by the carpet?

Hint: Maximum occurs at $x = 1$ and maximum value is $\int_1^4 t(5 - t)dt$



Thank You...

